



TrueVision Software Design

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Brief Summary - Problem

- There are a lot of AI Images
- Sometimes we need to know if images are AI
- How do tell if an image is AI





Brief Summary - Solution

- Have AI determine if an image is AI
- Put our AI in a web service
- Offer our model as an API



Changes to Requirements

2 Additions and 1 Revision





The Addition

Req 4-6: The results page shall allow users to view an AI attention heatmap overlay over the analysed image if they are a paid user

- **Req 4-6.1:** The results page shall allow users to turn off and on the heatmap overlay



The Revision

Req 4-2: The page shall allow users to upload another image to analyze

- **Req 4-2.1:** The page shall allow users to upload an image from their device
- **Req 4-2.2:** The page shall allow users to input a URL to an image

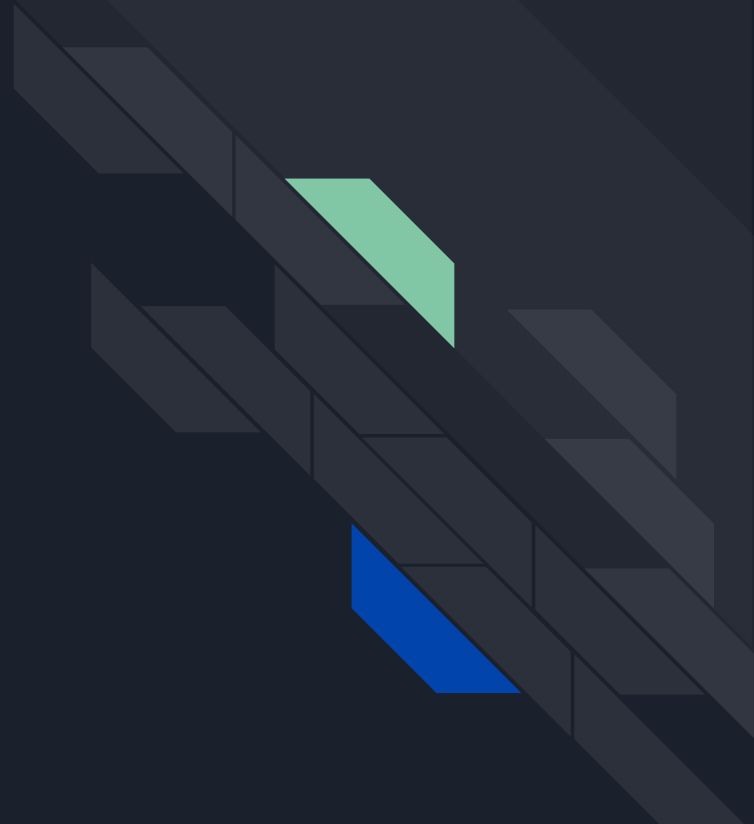
Req 4-3: The page shall allow users to click to confirm their image upload and begin analysis



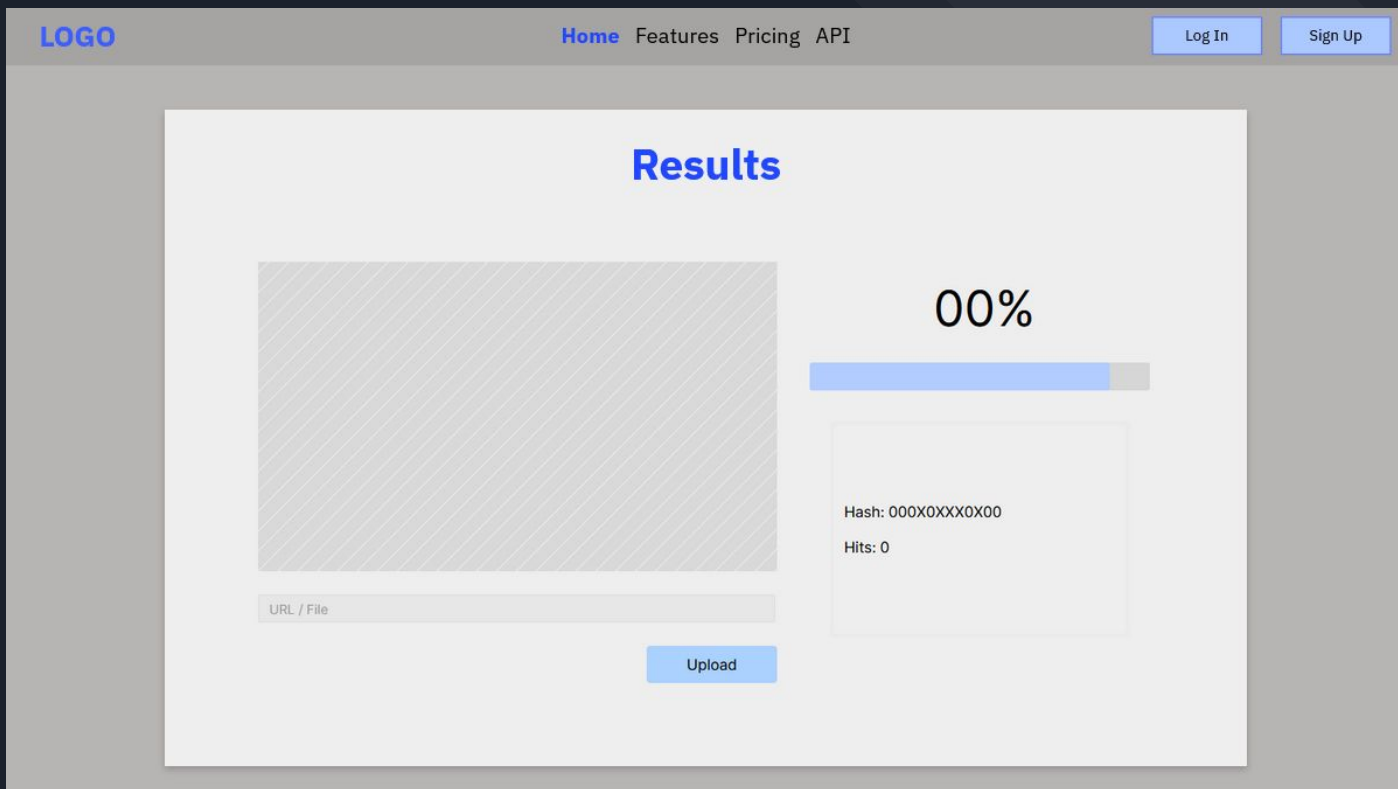
Req 4-2: The page shall link back to the home/upload page

Changes to Design?

Yes



Result Page Previously



Result Page Now

[TrueVision](#)[Analyze](#)[Features](#)[Pricing](#)[API](#)[Log In](#)[Sign Up](#)

Model Used
art

Our analysis says this image is:
Human

93.4%
Confidence

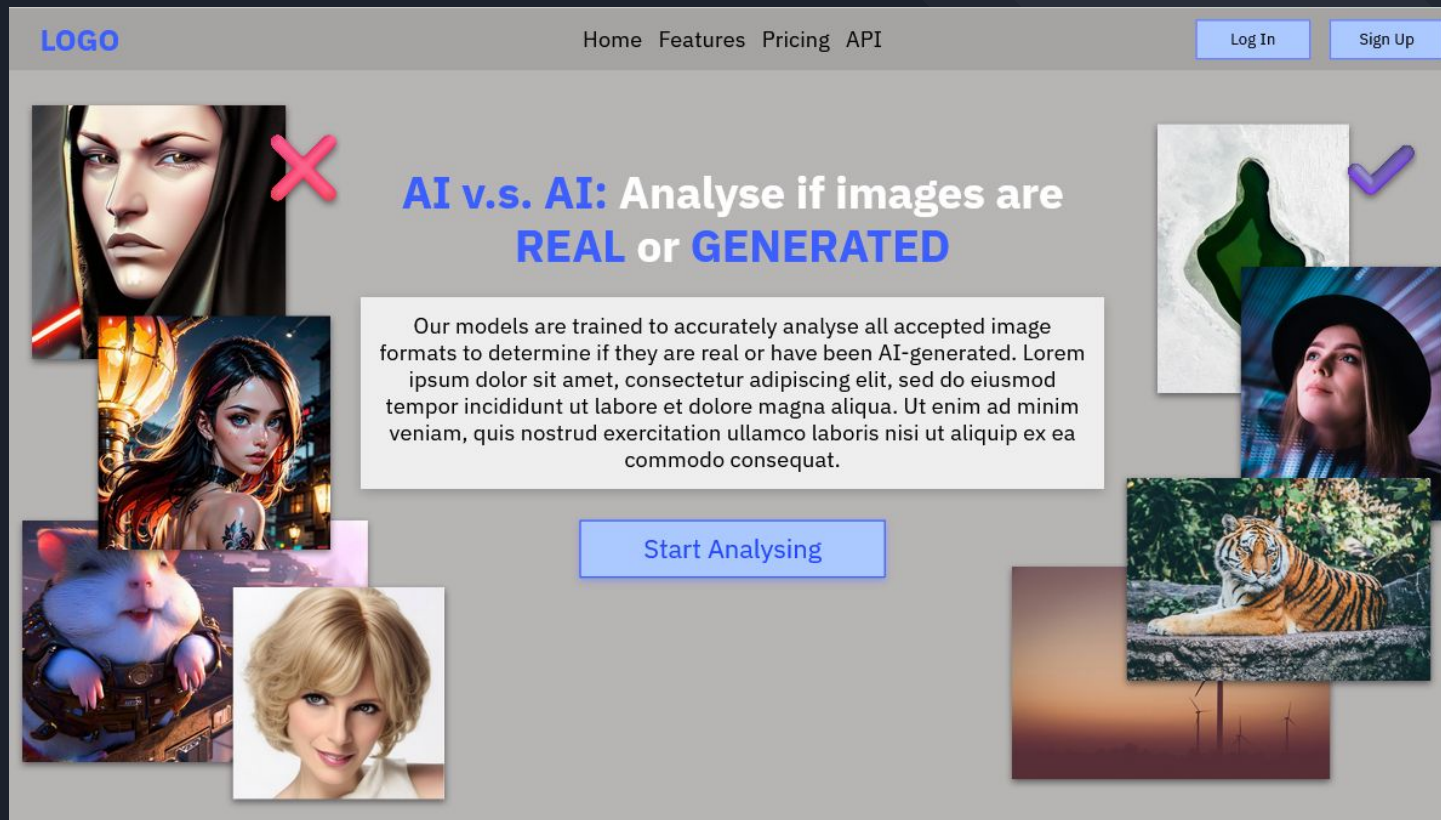
Scan Another Image

Model's Attention

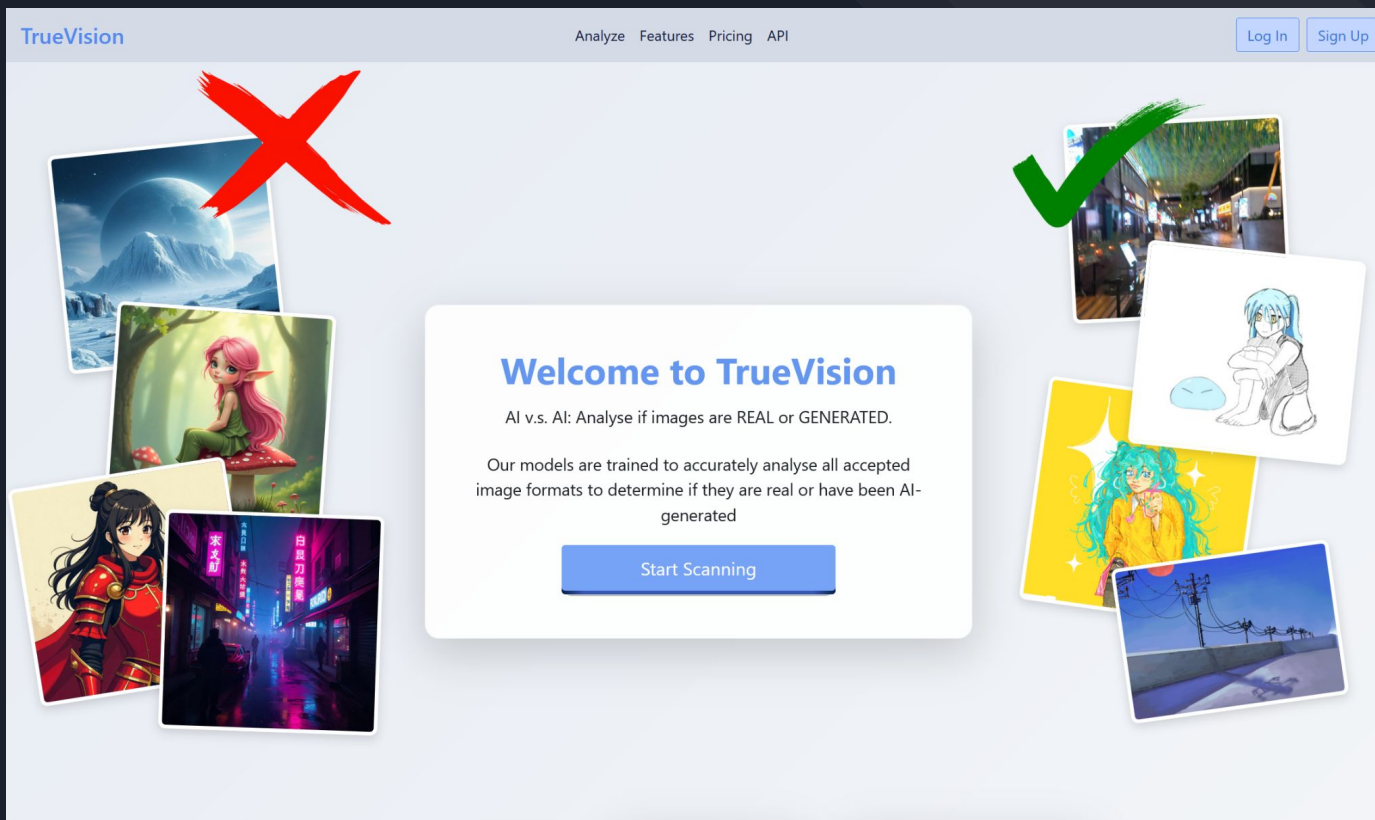


Display Options
☒ Show Overlay
Swap to Mask
When overlay is off, the original image is displayed.

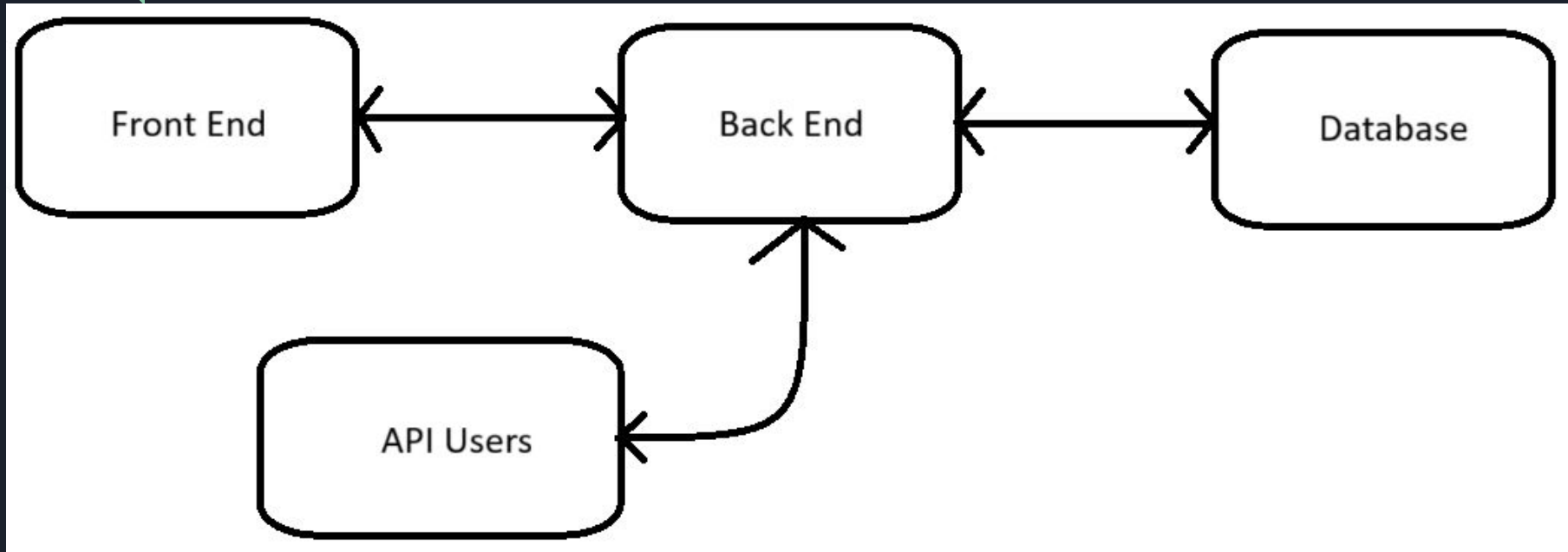
Landing Page Previously



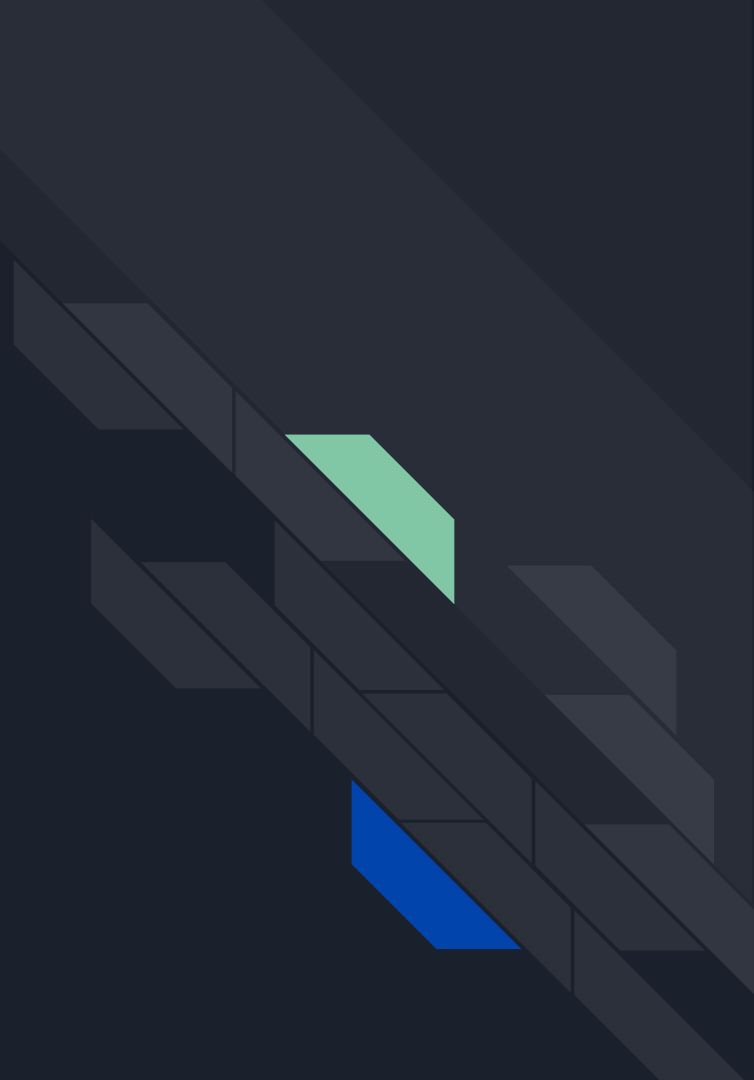
Landing Page Now - no more copyright infringement



Architectural Design



Chosen Technologies





Front End

Chosen Technologies:

- Github Pages
- React + Bootstrap
- Typescript

Reasons:

- Free
- Provides HTTPS
- Streamlines Development
- Catches Errors Early



Back End

Chosen Technologies & Libraries:

- Personal Desktop
- Python
- FastAPI
- Transformers:
vit-base-patch16-224-in21k

Reasons:

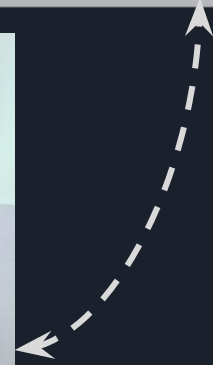
- Free (Technically)
- Python is a simple language
- Possible to Create Custom Models



Back End - Database

Chosen Libraries:

- PyMongo
 - Python driver for MongoDB
 - Compatible with FastAPI and async calls
- PyDantic
 - Automatically converts between BSON (MongoDB) and JSON (backend)
 - Ensures type and collection data validation





Database

Chosen Technologies:

- MongoDB
- MongoDB Atlas cloud service
 - 3 AWS instances

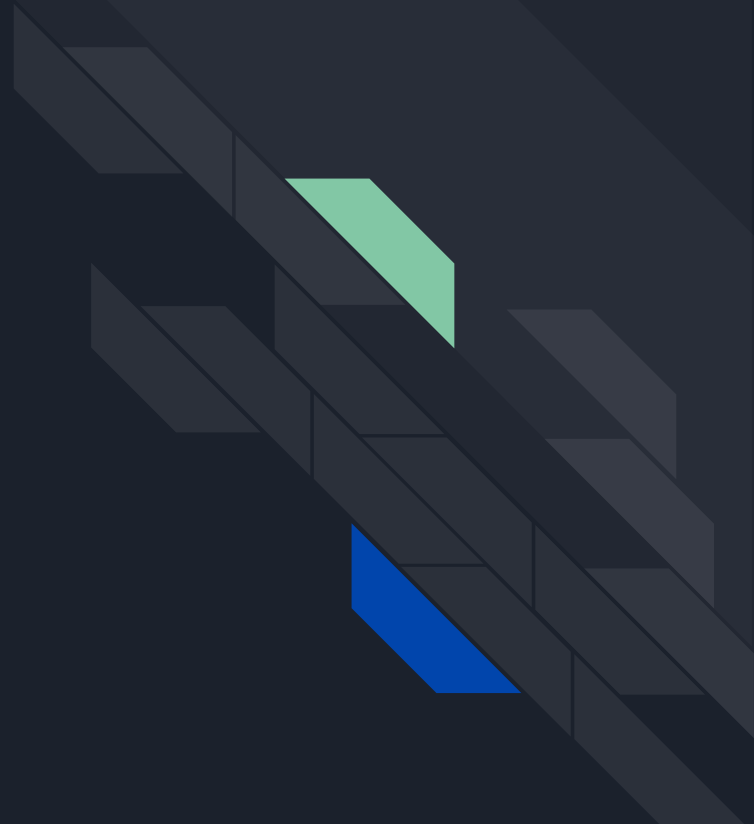
Reasons:

- Free
- Adequate storage and performance
- Flexible schema
- Auto generated ObjectID

The AWS logo is shown in the bottom right corner. It consists of the letters "AWS" in a bold, white, sans-serif font, set against a background of horizontal orange and white stripes.

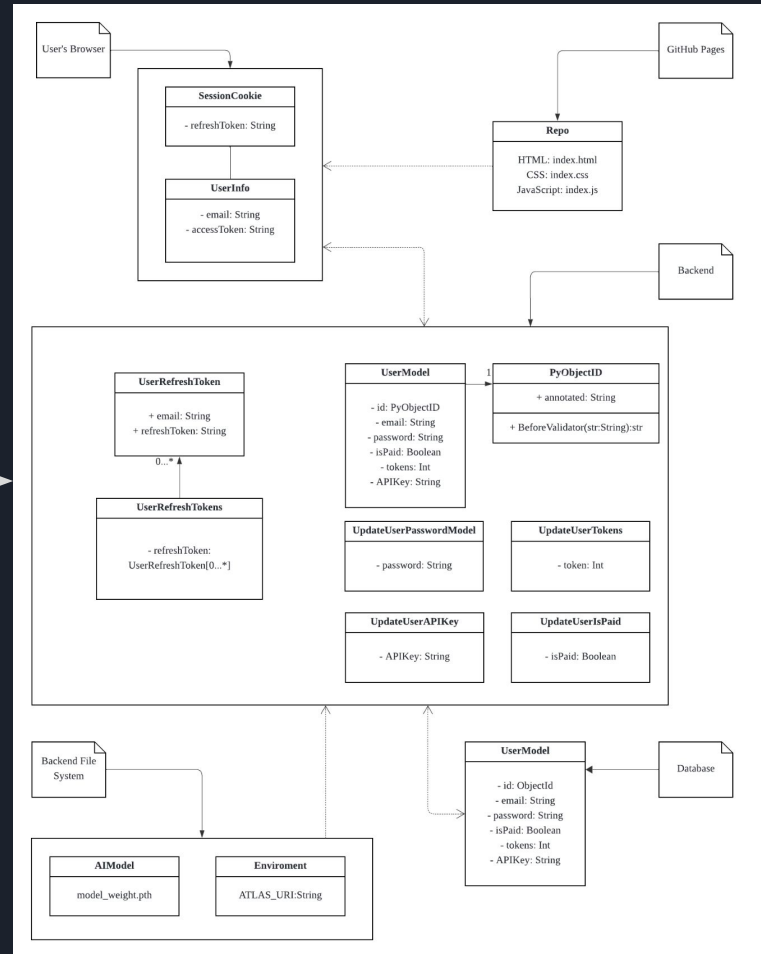
AWS

Data Design

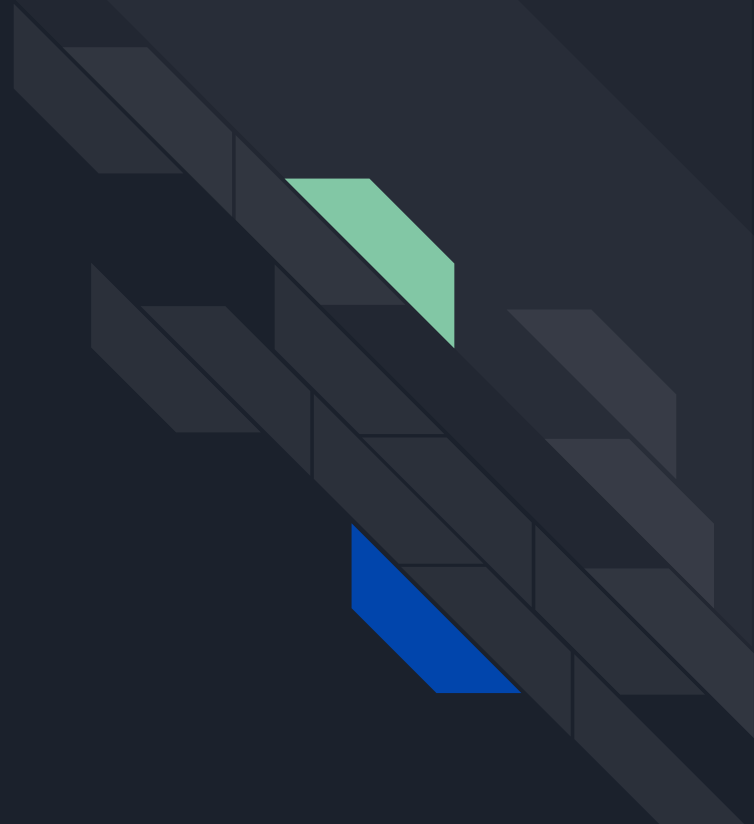


Data Design

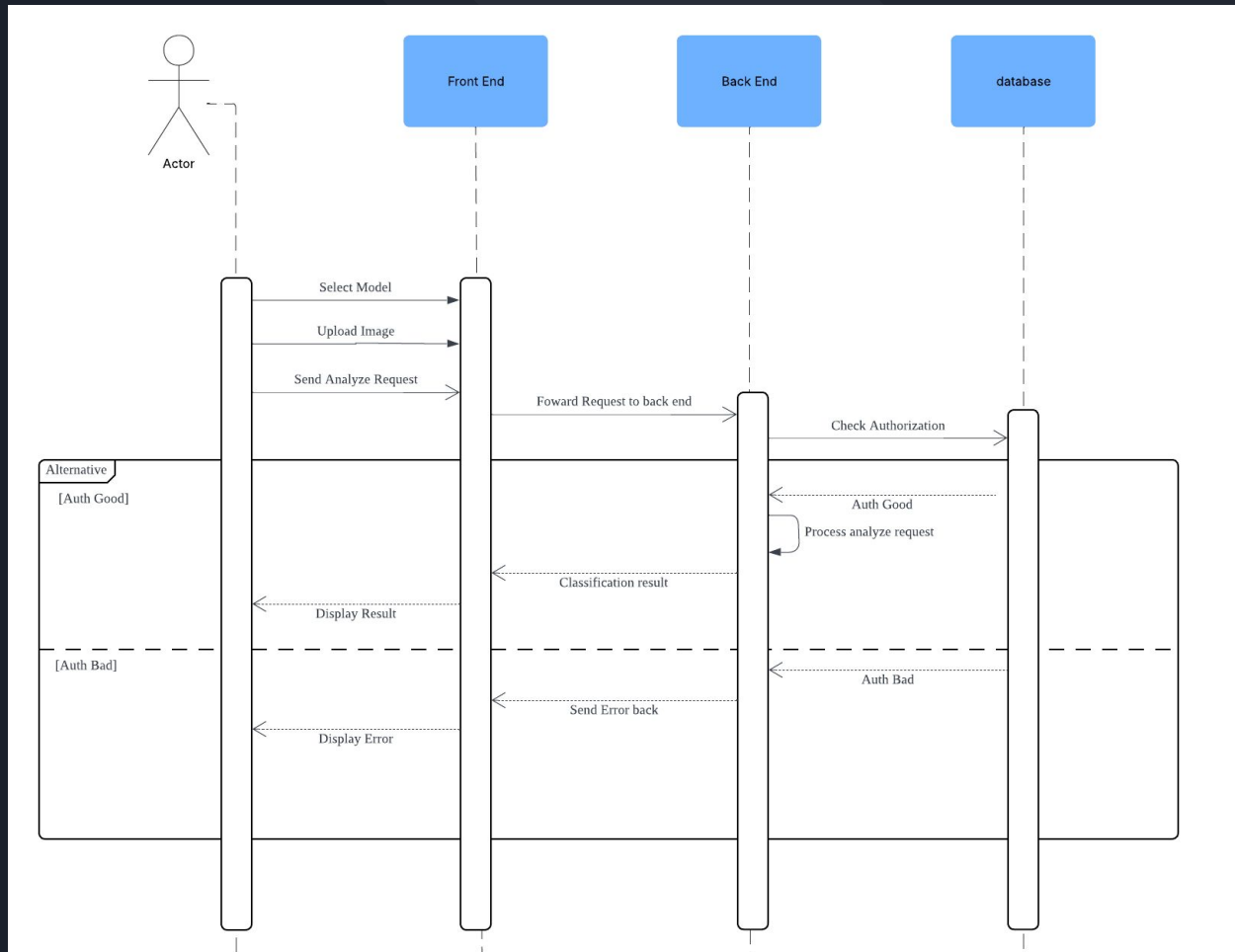
Reminder



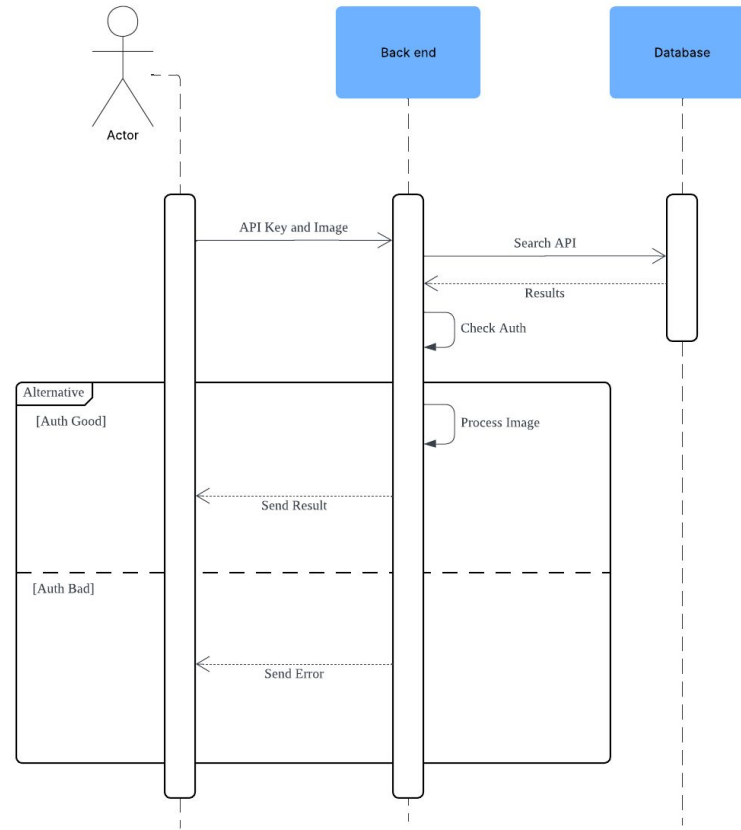
Data Sequence Diagrams



User Uses Analyze Image Feature



Advanced User API Call



Schedule



Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
27	28	29	30 Milestone 1	31	1	2
3	4 Milestone 2	5	6	7	8	9
10	11 Milestone 3	12	13	14	15	16
17	18	19	20 Milestone 4: Beta Release	21	22	23

Milestone 1: 30th October

- Minsu: Frontend API functionality for user login, signup, and subscription purchase, Complete Purchase page and link as required
- Andy: Complete token Purchase page, Adjust frontend API calls on Analyze page

Milestone 2: 4th September

- Minsu: Add authentication and database checks to login, signup, change password, purchase plan and token, cancel plan, add delete account API calls
- Andy: Add key checking to non-frontend Analyze Image API calls, Make API token generator API, Train Anime Art Model

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Milestone 3: 11th September

- Test and debug frontend, backend, database, and API functionality
- Document and fix issues
- Deploy application for Beta Release

Milestone 4: 20th September

- Get user feedback on our website
- Implement any required changes from feedback and fix any bugs found
- If possible: Implement stretch goals (image hashing, Google auth, email verification, multithreaded backend)

Questions?

